

DSA STUDY BLUEPRINT SERIES

25. Strings Part4 Reverse Words

Reversing Words in a String In-Place

Section 04 • C++ Core Data Structures & Algorithms

Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **Problem:** Reverse character positions inside individual word boundaries.
- Identify start/end indices of words, then perform reverse swaps.

Memory & Execution Blueprint

"hello world" ⇒ "olleh dlrow"

Word boundaries identified; characters swapped in-place.

C++ Implementation Reference

Standard code layout and syntax

```
void reverseWords(string &s) {  
    int n = s.length(), i = 0;  
    while (i < n) {  
        while (i < n && s[i] == ' ') i++; // skip spaces  
        int start = i;  
        while (i < n && s[i] != ' ') i++;  
        int end = i - 1;  
        if (start < n) reverse(s.begin() + start, s.begin() + end + 1);  
    }  
}
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
Time Complexity	$O(N)$
Space Complexity	$O(1)$ auxiliary

Key Practices & Gotchas

- **Important:** Can also be done using a stringstream buffer in C++ if extra space is permitted.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's dry run two-pointer string validations or window updates for string "race" :

Step	Left Pointer Index	Right Pointer Index	Char Comparison	Outcome
1	left = 0 ('r')	right = 3 ('e')	'r' != 'e'	Validation stops (Not a palindrome)

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Sliding Window Patterns:** Track character frequency maps within a moving boundary window size to find substring anagrams and permutations.
- **In-place reversals:** Swap characters inside word boundaries using double pointer offsets.
- **Related LeetCode Problems:** #3 Longest Substring Without Repeating Characters, #567 Permutation in String, #344 Reverse String.